

Aanchal Save

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EDUCATION

Georgia Institute of Technology (GT), Atlanta, GA

May 2028

Bachelor of Science in Aerospace Engineering - Minor: Computer Science

GPA: 4.0/4.0

EXPERIENCE

Palantir Technologies – *Software Engineering Intern*

Incoming Summer 2026

SpaceX – *Guidance, Navigation, and Controls (GNC) Software Engineering Intern*

Sept 2025 – Dec 2025

- Designed upcoming Starshield mega constellations to build future business and government architectures
- Helped inform trades on internally developed sensing payload, using simulations to create novel metrics and analysis tools
- Creating and improving physics-based constellation simulations and developing plot analysis tools using C++ and Python

Amentum Space Exploration Division – *ML and Robotics Intern at NASA MSFC (EV-43)*

May 2025 – Aug 2025

- Developed software at NASA Marshall Space Flight Center on a 7 degree of freedom robotic arm for lunar habitats
- Created deep learning architecture in PyTorch to solve kinematics problems using Gazebo and Simulink simulation data
- Accepted for NASA New Technology Report (NTR)** toward patent of one-of-a-kind, accurate inverse kinematic method
- Enabling natural language control with Large Language Models and RAG using ROS2 through distributed edge devices, interfacing with YOLO v11 object detection using LLaMA, Whisper, and Chainlit

Skyfall Space – *Flight Software Engineer*

May 2025 – Aug 2025

- NASA funded startup, eliminating space debris using CubeSats, creating GNC algorithms and simulating debris propagation

RoboJackets at GT – **RoboNav** – *Software and Mechanical Team – University Rover Challenge*

Aug 2024 – Present

- Using ROS2 and C++ for autonomous rover navigation, creating a traversability map from LIDAR data for motion planning
- Interfaced with path planning and mechanical team to understand traversability and shape processing of sensor inputs

RESEARCH

DREAM Lab at Georgia Tech Europe – *Undergraduate Research Assistant*

Jan 2026 – Present

- Implemented and evaluated TD-MPC2 world-model reinforcement learning, combining learned latent dynamics with online planning to achieve robust, data-efficient control on robotic navigation benchmarks and comparing with PPO and PID
- Developed cross-domain evaluation experiments in Nvidia Isaac Lab, Google Deepmind MuJoCo, and MetaWorld, analyzing the scalability of next-generation model-based autonomy algorithms and conducted sim-to-real robustness studies

NASA Green Propulsion Dual Mode (GPDM) Mission – *Flight Operations Team Member*

Feb 2025 – Aug 2025

- GT's Space Systems Design Laboratory (SSDL) with NASA MSFC, satellite mission integration, testing, and operations
- Using Orekit and Python to write navigation software for orbit propagation and thrust estimation
- Quality assurance for integration and testing of subsystems in cleanroom, including 'day-in-the-life' tests for flight readiness

CEREAL Lab at GT – *Undergraduate Research Assistant*

Nov 2024 – May 2025

- Computational and Experimental Rotorcraft Engineering and Aerodynamics Lab, research sponsored by the US Army
- Developing Bayesian machine learning (ML) frameworks for predicting wing pressure distribution using historic airfoil data
- Exploring applications of large language models (LLMs) and RAG to deliver natural language insights on aerodynamics

Dynamics and Controls Systems Laboratory at GT – *Undergraduate Research Assistant*

Aug 2024 – Dec 2024

- Implemented object detection and avoidance with YOLO and disparity-based depth estimation for autonomous navigation on an F1-Tenth platform with LIDAR and a stereo camera, using ROS2 and C++. Research sponsored by Sandia National Labs

PUBLICATIONS

Conference Presentation and Publication - A. Save, Artificial Neural Networks for Solving Forward and Inverse Kinematics on a 7 Degree-of-Freedom Robotic Manipulator. IEEE ICMRE 2026 (International Conference on Mechatronics and Robotics Engineering) Accepted for oral presentation and publication in IEEE Xplore.

Conference Presentation and Publication – W. Lowman, A. Save, D. Ho et al., A Distributed Intelligence Architecture for Responsive Command Handling in Edge Robotics, AIAA ASCEND 2026. Accepted for oral presentation and publication.

Journal Publication - H. Lee, A. Save, P. Seshadri et al., Large airfoil models. Computers and Fluids (2025), doi: <https://doi.org/10.1016/j.compfluid.2025.106662>.

Conference Publication - Exploring Atmospheric Effects During Solar Eclipses: High-Altitude Ballooning Data Collection & Analysis. American Geophysical Union Fall Meeting 2023

SKILLS

Certifications: Certified Solidworks Associate, Java IT Specialist, Autodesk Inventor, Microsoft Office Specialist

Software: Linux, ROS 2, Gazebo, Android Studio, Unity, Blender, SQL

Programming Languages: Java, Python, MATLAB, C++, C#, SQL Script, HTML, Javascript

Hardware and Machining: Embedded Systems, Arduino, Raspberry Pi, Mill, Lathe, CNC, Waterjet